

Agentic AI Framework for Business & Finance Intelligence

 launch  binder  Open in Colab

No install needed to try it yourself — click either badge above to get a live, runnable copy of this exact notebook in your browser. Binder needs no account at all; Colab needs a free Google login. Both run the real pipeline against live news.

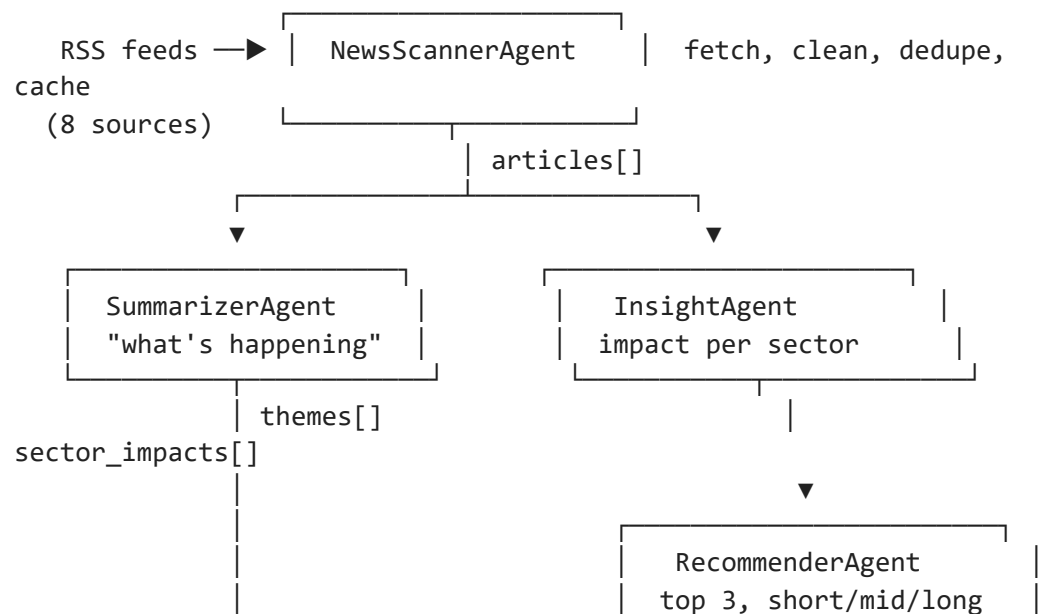
Internship Task 2 — an agentic pipeline that:

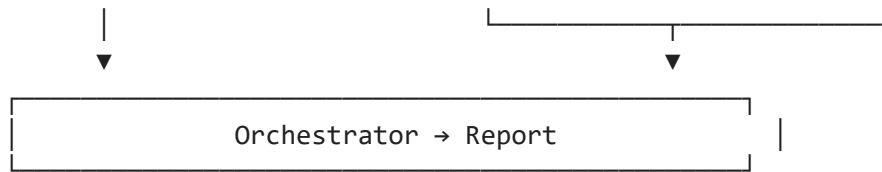
#	Requirement	Agent responsible
a	Scan multiple news sources and summarize what's happening (Business & Finance POV)	NewsScannerAgent → SummarizerAgent
b	Generate insights and assess impact on major industries	InsightAgent
c	Recommend the top 3 sectors to focus on, Short/Mid/Long-term	RecommenderAgent

Each requirement maps to its own agent. An **Orchestrator** sequences them and passes state between them — that hand-off between autonomous, single-purpose agents is what makes this *agentic*, rather than one long prompt asking an LLM to "do everything."

Run this notebook top to bottom to fetch live news and regenerate the full briefing.

Architecture





Two reasoning modes, same interface. Every agent first tries a real LLM call (Groq's free tier — an OpenAI-compatible API). If no key is set, or the call fails for any reason (network, rate limit, bad response), the agent transparently drops to a deterministic, keyword/lexicon-based fallback so the pipeline *never* crashes mid-demo. `LLMClient` is the only piece of code that talks to the network — every agent just asks it for JSON and gets either a result or `None`.

Grounding, not vibes. Every theme, every sector rating, every recommendation carries the specific article(s) it was derived from. Nothing in the final report is asserted without a clickable source.

```
In [1]: import sys, os

# Colab only fetches this single .ipynb, not the rest of the repo – the
# other agent files it imports below live alongside it on GitHub, so on
# Colab we clone the repo first. No-op on Binder/local, since there the
# full repo is already on disk.
if "google.colab" in sys.modules:
    REPO_DIR = "agentic-ai-framework"
    if not os.path.exists(REPO_DIR):
        !git clone -q https://github.com/rkcapitalx953-sys/Rounak-Agentic-AI-Framework
    %pip install -q -r {REPO_DIR}/requirements.txt
    os.chdir(REPO_DIR)
```

```
In [2]: import sys, time
from pathlib import Path

# Make sure the project root (this notebook's folder) is importable
sys.path.insert(0, str(Path.cwd()))

from dotenv import load_dotenv
load_dotenv()

import html as html_lib
from collections import Counter

import config
from orchestrator import Orchestrator
from report import render_markdown, render_sector_chart

import matplotlib.pyplot as plt
from IPython.display import display, Markdown, Image, HTML

def show_table(rows, columns):
    # small dependency-free table renderer (no pandas required)
```

```

head = "".join(f"<th style='text-align:left;border-bottom:2px solid #999;padding:
body = ""
for row in rows:
    cells = "".join(f"<td style='padding:4px 12px;border-bottom:1px solid #eee'
    body += f"<tr>{cells}</tr>"
display(HTML(f"<table style='border-collapse:collapse'><tr>{head}</tr>{body}</tr>"))

```

```

In [3]: orchestrator = Orchestrator()

if orchestrator.llm.available:
    print(f"LLM reasoning ENABLED – model: {orchestrator.llm.model}")
else:
    print("No GROQ_API_KEY found – running on deterministic fallback agents.")
    print("Get a free key (no card) at https://console.groq.com/keys, put it in a .env file")
    print("as GROQ_API_KEY=..., then re-run this cell for the full LLM-reasoned experience.")

```

No GROQ_API_KEY found – running on deterministic fallback agents.
 Get a free key (no card) at <https://console.groq.com/keys>, put it in a .env file
 as GROQ_API_KEY=..., then re-run this cell for the full LLM-reasoned experience.

Run the pipeline

```

In [4]: t0 = time.time()
result = orchestrator.run()
elapsed = time.time() - t0

print(f"Pipeline finished in {elapsed:.1f}s")
print(f"Articles analyzed : {result.meta['article_count']}")
print(f"Reasoning mode      : {'LLM (Groq)' if result.meta['llm_available'] else 'Rule-based fallback'}")
print(f"Served from cache   : {result.meta['used_cache']}")
if result.meta['source_errors']:
    print(f"Sources that failed this run: {list(result.meta['source_errors'].keys())}")

```

Pipeline finished in 7.0s
 Articles analyzed : 60
 Reasoning mode : Rule-based fallback
 Served from cache : False

Agent 1 — NewsScannerAgent

Fetches 8 independent public RSS feeds (wire services, business press, markets desks) in parallel-ish fashion, strips HTML, de-duplicates near-identical wire stories by normalized title, and caches the result to disk so a flaky connection during a live demo doesn't kill the run.

```

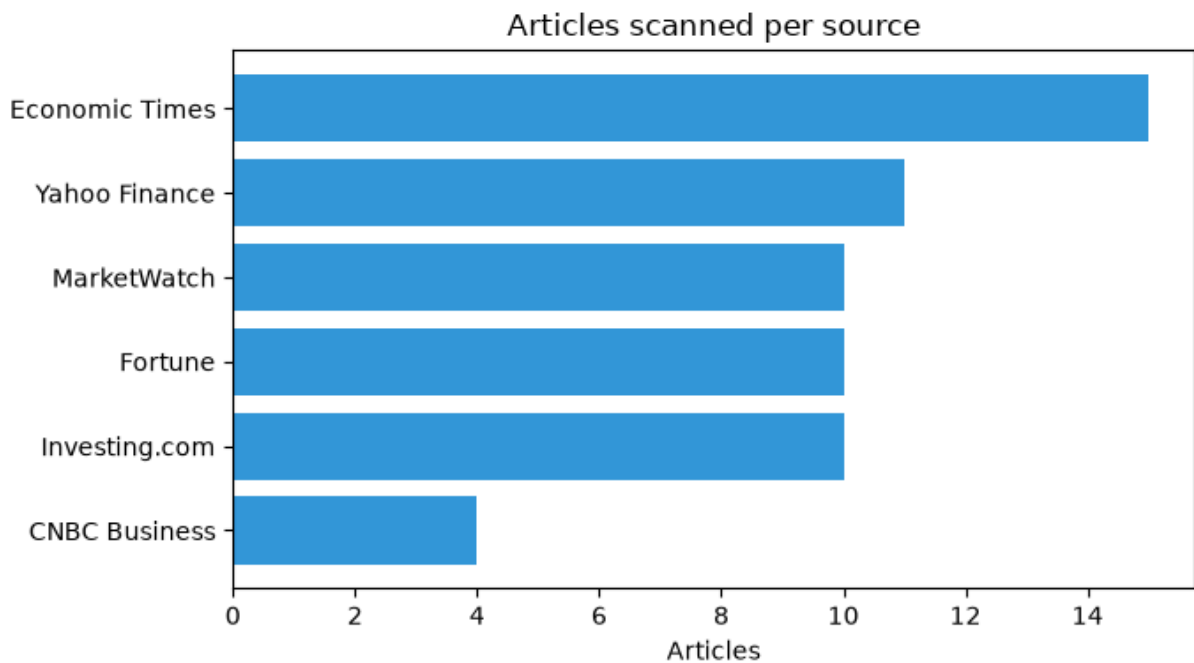
In [5]: show_table(
    [[a["source"], a["title"], a["published_iso"]] for a in result.articles[:10]],
    ["Source", "Title", "Published"],
)

counts = Counter(a["source"] for a in result.articles)
sources, freqs = zip(*sorted(counts.items(), key=lambda kv: kv[1]))
plt.figure(figsize=(7, 4))

```

```
plt.barh(sources, freqs, color="#3498db")
plt.xlabel("Articles")
plt.title("Articles scanned per source")
plt.tight_layout()
plt.show()
```

Source	Title	Published
Investing.com	PostNL H1 free cash flow beats, keeps 2026 outlook despite volume weakness	2026-08-03T00:26:59+00:00
Investing.com	WeRide expands to Denmark in continued European market push	2026-08-03T00:25:53+00:00
Investing.com	Gold rises as weaker dollar offsets Fed concerns	2026-08-03T00:22:44+00:00
Investing.com	Stabilus Q3 profit jumps on disposal gains, narrows FY outlook	2026-08-03T00:19:47+00:00
Investing.com	Kay Granger, first Republican woman to lead House Appropriations Committee, dies at 83	2026-08-03T00:19:38+00:00
Investing.com	Shein weighs cost reset for late-stage investors ahead of Hong Kong IPO, Bloomberg News reports	2026-08-03T00:19:21+00:00
Investing.com	184%+ gains and counting: A new list of AI-picked stocks for August IS NOW LIVE	2026-08-03T00:19:17+00:00
Investing.com	Yen jumps as traders watch for further intervention	2026-08-03T00:19:00+00:00
Investing.com	Sandoz agrees \$450 million settlement with 43 US states over antitrust claims	2026-08-03T00:18:36+00:00
Investing.com	Jyske Bank buys back shares worth DKK 58 million in week 31	2026-08-03T00:12:34+00:00



Agent 2 — SummarizerAgent — Requirement (a): what's happening

Reads the scanned headlines and writes a themed "what's happening" briefing. On the LLM path, the model is given an *indexed* headline list and must cite which indices support each theme — indices are then mapped back to real article links, so every paragraph below is traceable to a source.

```
In [6]: for theme in result.themes:
        md_block = f"### {theme['headline']}\n\n{theme['narrative']}\n\n"
        for a in theme["articles"][:3]:
            md_block += f"- [{a['title']}]({a['link']}) - *{a['source']}*\\n"
        display(Markdown(md_block))
```

Financials

Gold rises as weaker dollar offsets Fed concerns. Jyske Bank buys back shares worth DKK 58 million in week 31. Banks propose 'Yes-No' for high-risk UPI transfers.

- [Gold rises as weaker dollar offsets Fed concerns](#) — *Investing.com*
- [Jyske Bank buys back shares worth DKK 58 million in week 31](#) — *Investing.com*
- [Banks propose 'Yes-No' for high-risk UPI transfers](#) — *Economic Times*

Information Technology

CXMT's blockbuster IPO will test whether China's memory makers are ready for the spotlight: 'It does not yet mean China is broadly catching up'. AI is creating a two-speed jobs market in the UK, Indeed says. As AI creates more entrepreneurs and a skills mismatch, Gen Z faces the choice of being a specialist or a 'general-purpose nerd'.

- [CXMT's blockbuster IPO will test whether China's memory makers are ready for the spotlight: 'It does not yet mean China is broadly catching up'](#) — *Fortune*
- [AI is creating a two-speed jobs market in the UK, Indeed says](#) — *Fortune*
- [As AI creates more entrepreneurs and a skills mismatch, Gen Z faces the choice of being a specialist or a 'general-purpose nerd'](#) — *Fortune*

General Business

PostNL H1 free cash flow beats, keeps 2026 outlook despite volume weakness. WeRide expands to Denmark in continued European market push. Stabilus Q3 profit jumps on disposal gains, narrows FY outlook.

- [PostNL H1 free cash flow beats, keeps 2026 outlook despite volume weakness](#) — *Investing.com*
- [WeRide expands to Denmark in continued European market push](#) — *Investing.com*
- [Stabilus Q3 profit jumps on disposal gains, narrows FY outlook](#) — *Investing.com*

Energy

Here's how Trump has zig zagged with on-again, off-again strikes throughout the Iran war. Suriname's \$26 Billion Oil Bet Is Finally Paying Off.

- [Here's how Trump has zig zagged with on-again, off-again strikes throughout the Iran war](#) — *Fortune*
- [Suriname's \\$26 Billion Oil Bet Is Finally Paying Off](#) — *Yahoo Finance*

Materials

Jindal Steel cuts capex, focuses on capacity use.

- [Jindal Steel cuts capex, focuses on capacity use](#) — *Economic Times*

Industrials

Why flights are so expensive and will likely stay that way.

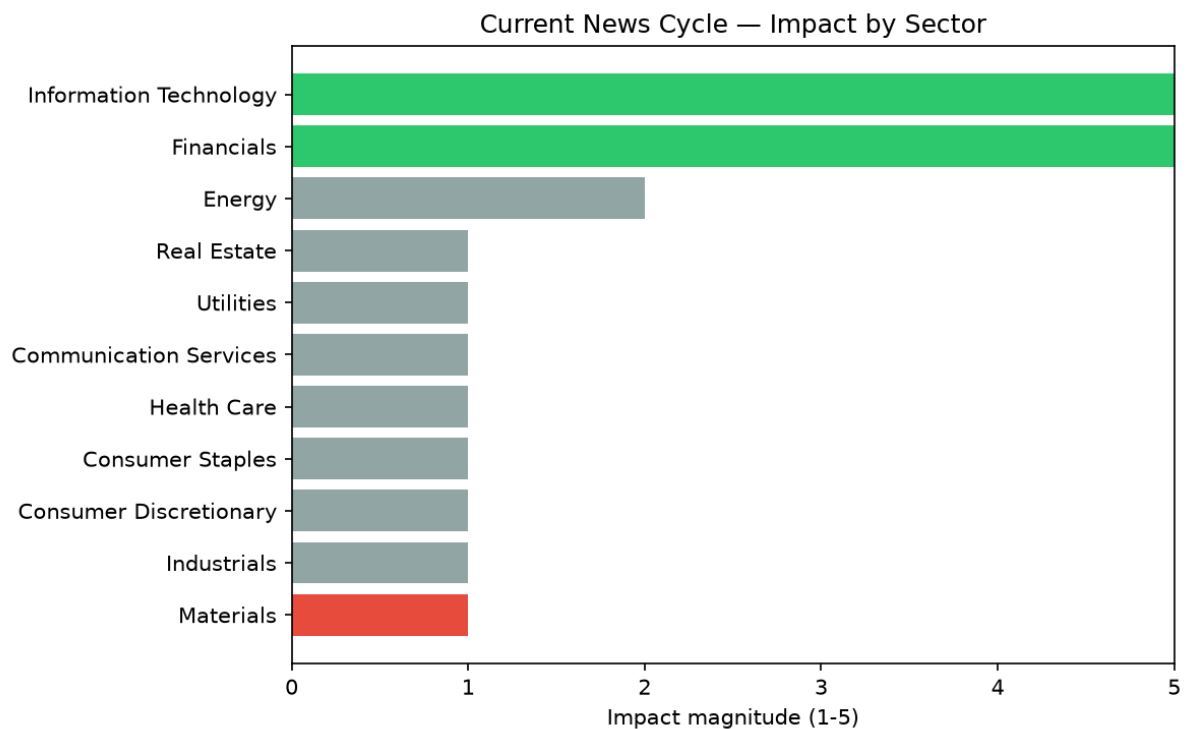
- [Why flights are so expensive and will likely stay that way](#) — *CNBC Business*

Agent 3 — InsightAgent — Requirement (b): industry impact assessment

Every one of the 11 GICS-style sectors gets a structured verdict: direction (Positive / Negative / Mixed / Neutral), a 1-5 impact magnitude, a grounded rationale, and supporting articles.

```
In [7]: show_table(  
        [[s["sector"], s["direction"], s["magnitude"], s["rationale"]] for s in result.  
         ["Sector", "Direction", "Magnitude", "Rationale"]],  
        )  
  
chart_path = str(config.REPORTS_DIR / "sector_impact_notebook.png")  
render_sector_chart(result, chart_path)  
display(Image(chart_path))
```

Sector	Direction	Magnitude	Rationale
Financials	Positive	5	7 related headline(s) this cycle (3 positive vs 0 negative signal words detected).
Information Technology	Positive	5	7 related headline(s) this cycle (4 positive vs 0 negative signal words detected).
Energy	Neutral	2	2 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Materials	Negative	1	1 related headline(s) this cycle (0 positive vs 1 negative signal words detected).
Industrials	Neutral	1	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Consumer Discretionary	Neutral	1	No significant coverage found in this news window.
Consumer Staples	Neutral	1	No significant coverage found in this news window.
Health Care	Neutral	1	No significant coverage found in this news window.
Communication Services	Neutral	1	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Utilities	Neutral	1	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Real Estate	Neutral	1	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).



Agent 4 — RecommenderAgent — Requirement (c): top 3 sectors, Short/Mid/Long-term

`RecommenderAgent` never re-reads raw headlines — it only sees the *already assessed* sector impacts from `InsightAgent`. Each agent has a narrow input contract instead of every agent redoing everyone else's work; that's the other half of what makes this "agentic" rather than one monolithic prompt.

```
In [8]: for i, pick in enumerate(result.recommendations, start=1):
        block = f"### {i}. {pick['sector']} · Conviction: {pick.get('conviction', 'Med')}\n"
        block += f"- **Short-term:** {pick['short_term']}\n"
        block += f"- **Mid-term:** {pick['mid_term']}\n"
        block += f"- **Long-term:** {pick['long_term']}\n"
        for a in pick.get("articles", []):2:
            block += f" - Evidence: [{a['title']}]({a['link']})\n"
        display(Markdown(block))
```

1. Financials · Conviction: High

- **Short-term:** Immediate catalyst: 7 related headline(s) this cycle (3 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
- **Mid-term:** Over the next 2-6 months, direction depends on whether the current driver persists rather than fades as a one-off news cycle.
- **Long-term:** 12+ months out, treat this as a secondary factor — the structural case for Financials should rest on fundamentals beyond this single news window.
 - Evidence: [Gold rises as weaker dollar offsets Fed concerns](#)
 - Evidence: [Lyske Bank buys back shares worth DKK 58 million in week 31](#)

2. Information Technology · Conviction: High

- **Short-term:** Immediate catalyst: 7 related headline(s) this cycle (4 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
- **Mid-term:** Over the next 2-6 months, direction depends on whether the current driver persists rather than fades as a one-off news cycle.
- **Long-term:** 12+ months out, treat this as a secondary factor — the structural case for Information Technology should rest on fundamentals beyond this single news window.
 - Evidence: [CXMT's blockbuster IPO will test whether China's memory makers are ready for the spotlight: 'It does not yet mean China is broadly catching up'](#)
 - Evidence: [AI is creating a two-speed jobs market in the UK, Indeed says](#)

3. Energy · Conviction: Medium

- **Short-term:** Immediate catalyst: 2 related headline(s) this cycle (0 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
- **Mid-term:** Over the next 2-6 months, direction depends on whether the current driver persists rather than fades as a one-off news cycle.
- **Long-term:** 12+ months out, treat this as a secondary factor — the structural case for Energy should rest on fundamentals beyond this single news window.
 - Evidence: [Here's how Trump has zig zagged with on-again, off-again strikes throughout the Iran war](#)
 - Evidence: [Suriname's \\$26 Billion Oil Bet Is Finally Paying Off](#)

Full report (auto-generated, ready to submit)

```
In [9]: report_md = render_markdown(result)
display(Markdown(report_md))
```

Business & Finance Intelligence Briefing

Generated 2026-08-03T17:21:48.317308+00:00 · 60 articles analyzed

Reasoning mode: Rule-based fallback (no LLM key set)

a) What's happening — Business & Finance briefing

Financials

Gold rises as weaker dollar offsets Fed concerns. Jyske Bank buys back shares worth DKK 58 million in week 31. Banks propose 'Yes-No' for high-risk UPI transfers.

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General Business

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- [Here's how Trump has zig zagged with on-again, off-again strikes throughout the Iran war](#) — *Fortune*
- [Suriname's \\$26 Billion Oil Bet Is Finally Paying Off](#) — *Yahoo Finance*

Materials

Jindal Steel cuts capex, focuses on capacity use.

- [Jindal Steel cuts capex, focuses on capacity use](#) — *Economic Times*

Industrials

Why flights are so expensive and will likely stay that way.

- [Why flights are so expensive and will likely stay that way](#) — *CNBC Business*

b) Industry impact assessment

Sector	Direction	Impact	Rationale
Financials	Positive	●●●●●	7 related headline(s) this cycle (3 positive vs 0 negative signal words detected).
Information Technology	Positive	●●●●●	7 related headline(s) this cycle (4 positive vs 0 negative signal words detected).
Energy	Neutral	●●○○○	2 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Materials	Negative	●○○○○	1 related headline(s) this cycle (0 positive vs 1 negative signal words detected).
Industrials	Neutral	●○○○○	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Consumer Discretionary	Neutral	●○○○○	No significant coverage found in this news window.
Consumer Staples	Neutral	●○○○○	No significant coverage found in this news window.
Health Care	Neutral	●○○○○	No significant coverage found in this news window.
Communication Services	Neutral	●○○○○	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Utilities	Neutral	●○○○○	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).
Real Estate	Neutral	●○○○○	1 related headline(s) this cycle (0 positive vs 0 negative signal words detected).

c) Top 3 sectors to focus on

1. Financials · Conviction: High

- **Short-term:** Immediate catalyst: 7 related headline(s) this cycle (3 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
- **Mid-term:** Over the next 2-6 months, direction depends on whether the current driver persists rather than fades as a one-off news cycle.
- **Long-term:** 12+ months out, treat this as a secondary factor — the structural case for Financials should rest on fundamentals beyond this single news window.
 - Evidence: [Gold rises as weaker dollar offsets Fed concerns](#)
 - Evidence: [Jyske Bank buys back shares worth DKK 58 million in week 31](#)

2. Information Technology · Conviction: High

- **Short-term:** Immediate catalyst: 7 related headline(s) this cycle (4 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
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3. Energy · Conviction: Medium

- **Short-term:** Immediate catalyst: 2 related headline(s) this cycle (0 positive vs 0 negative signal words detected). Expect this to shape sentiment over the next 2-4 weeks.
- **Mid-term:** Over the next 2-6 months, direction depends on whether the current driver persists rather than fades as a one-off news cycle.
- **Long-term:** 12+ months out, treat this as a secondary factor — the structural case for Energy should rest on fundamentals beyond this single news window.
 - Evidence: [Here's how Trump has zig zagged with on-again, off-again strikes throughout the Iran war](#)
 - Evidence: [Suriname's \\$26 Billion Oil Bet Is Finally Paying Off](#)

```
In [10]: from datetime import datetime
timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
report_path = config.REPORTS_DIR / f"briefing_{timestamp}.md"
report_path.write_text(report_md, encoding="utf-8")
print(f"Saved: {report_path}")
```

Saved: C:\Users\rounak\OneDrive - GEMS Education\Documents\AI-Financial-Intelligence\reports\briefing_20260803_225148.md

Design notes

- **1:1 traceability to the brief.** Four agents map directly onto requirements (a), (a), (b), (c) — easy for a grader to verify nothing was hand-waved.
- **Grounded, not hallucinated.** Every theme / sector rating / recommendation cites specific article links pulled from the actual scan, not invented by the model.
- **Graceful degradation.** No API key, a dead feed, or a network blip during the live demo does not crash the pipeline — it silently drops to an interpretable, keyword-lexicon fallback (see `config.SECTOR_KEYWORDS` , `config.POSITIVE_WORDS` / `NEGATIVE_WORDS`).
- **Free and reproducible.** No paid API required — a free Groq key (`console.groq.com` , no card) is enough to run the full LLM-reasoned pipeline anywhere, including for whoever grades this.

Appendix — running this yourself

Zero-install: click the Binder or Colab badge at the top of this notebook.

Locally:

```
pip install -r requirements.txt
cp .env.example .env          # add your free GROQ_API_KEY
jupyter notebook notebook.ipynb  # or: python main.py
```